

Project Topic

Your Name

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1 Introduction

The introduction will broadly introduce your topic, motivate it, and describe what will be done. Urban planners, public sector transportation employees, and politicians rely on accurate traffic accident data to make informed decisions that can improve traffic safety. This study aims to provide better insight into traffic accidents, including factors that can influence the likelihood of a traffic accident as well as its severity. Using XGBoost analysis, this study will analyze a dataset of 10,000 traffic accidents in Wooster, Ohio over the past five years in order to model the likelihood of accidents and determine which factors most contributed to causing a traffic accident. Severity of the traffic accident will additionally be considered.

2 Background

The field has historically relied on statistical regression analysis in order to make decisions regarding traffic safety, such as ARIMA or linear regression techniques. Recent approaches using machine learning and neural networks produce models of traffic flows, jams, and accidents have shown a marked increase in accurately modeling traffic. Signal traffic control is another field of study in which machine learning analysis is used to decrease congestion in urban settings.

Many studies have observed the superior performance of decision tree networks such as random forest and XGBoost over neural networks such as LSTM or CNNs. Urbanized cities in North America have more data available to train machine learning models or neural networks, such as Uber data, sensors, cameras, and satellite data. Developing cities face greater challenges in acquiring the data required to model traffic flows and often require collaboration with local governments in order to place sensors or otherwise acquire traffic data that can be used for analysis.

3 Related Works

4 Methodology

Dijkstra's algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [?].

5 Results

An exemplary figure is shown in Figure 1. Any time a figure is used, it must have a caption and be directly referred to and discussed in the text.

6 Conclusion

References

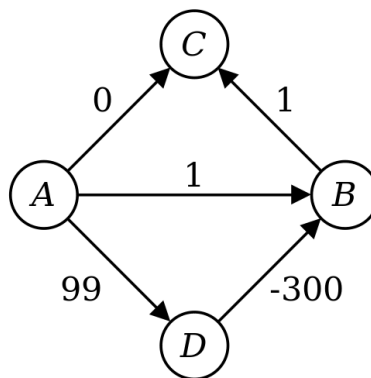


Figure 1: Dijkstra's algorithm does not work on graphs with negative edge weights, as evidenced here.